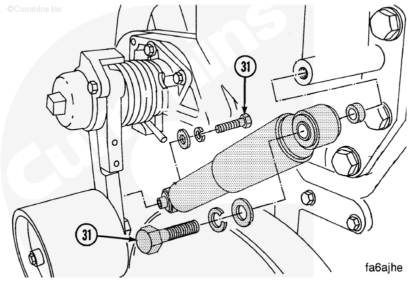
Trainings

Remove

Note : The back side idler system uses one of two types of control rods (turnbuckles) or a shock absorber. Refer to the OEM service manual for the instructions that apply to the engine being serviced.

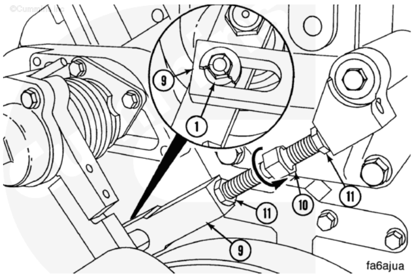
Remove the back side idler end of the shock absorber, solid control rod (turnbuckle), or control rod tensioner assembly.

Loosen the upper capscrew (31). Remove the lower capscrew (31).



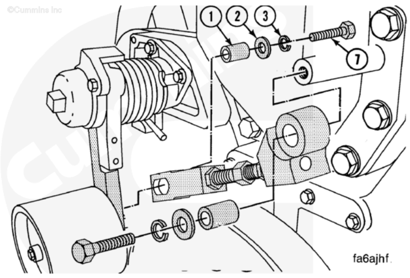
Note : One of the jam nuts on the solid control rod (turnbuckle) has left hand threads.

Loosen the solid control rod (turnbuckle) jam nuts (11). Turn the adjusting screw (10) until the spacer (1) is **not** touching the end of the slot in the control rod (9).

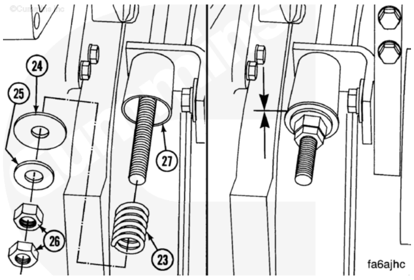


Remove the capscrew (7), washers (2)(3), and spacers (1).

Remove the control rod assembly from the idler assembly.

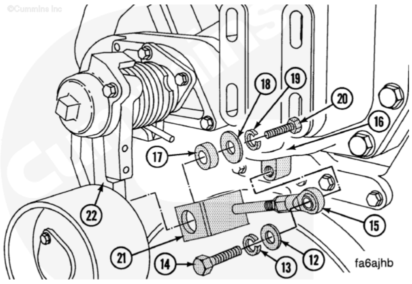


To remove the control rod with spring, remove the two jam nuts (26), washers (25)(24), and spring (23).



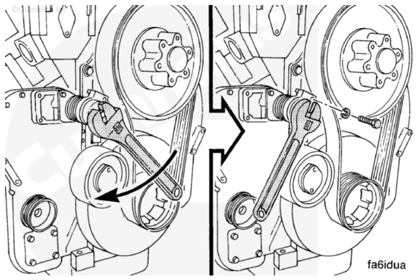
Remove the following parts:

- Capscrew (20)
- Lock washer (19)
- Washer (18)
- Spacer (17)
- Capscrew (14)
- Lock washer (13)
- Washer (12)
- Control rod (15).



⚠ WARNING ⚠

The fan belt idler is under tension. Do not allow hands to get between the idler, fan, or the fan hub. Failure to do so can result in personal injury.



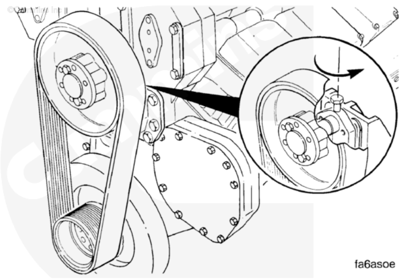
Use a 8-point socket and breaker bar or large wrench to hold the idler in position against the spring tension.

Remove the capscrews from the spring cap.

Slowly turn the wrench until the spring tension is relieved.

Remove the fan belt.

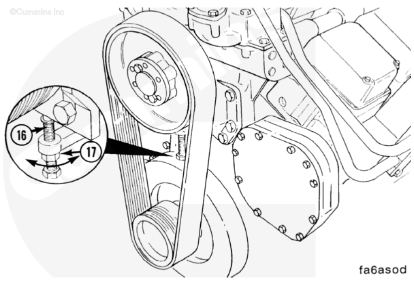
Note : The fan center distance is the distance between the crankshaft and fan center lines.



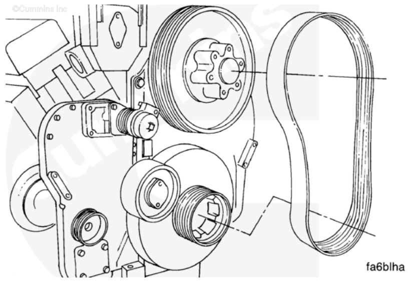
On systems that use a two pulley fan drive (without idler pulley) with a 508 mm [20 in], 559 mm [22 in], or 610 mm [24 in] fan center, loosen the bolts that pass through the slotted holes in the fan hub bracket.

Loosen the fan hub adjusting screw.

On engines that use a two pulley drive with a 711 mm [28 in] fan center distance, loosen the bolts that pass through the slotted holes and fan hub bracket of the 711 mm [28 in] fan center system. The adjusting screw (16) is below the fan hub. The locknut (17) **must** be loosened before loosening the fan hub adjusting screw.



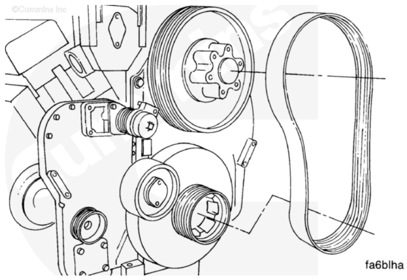
Remove the fan belt.



Install

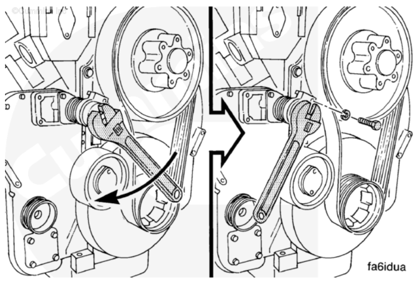
Install the belt on the crankshaft and fan hub pulley.

Align the grooves on the belt on the ribs of the pulley.



⚠ WARNING ⚠

The fan belt idler is under tension. Do not allow hands to get between the idler, fan, or the fan hub. Failure to do so can result in personal injury.



After installing the fan belt, install the back side fan idler system.

Rotate the idler against the spring tension until the capscrew holes are aligned.

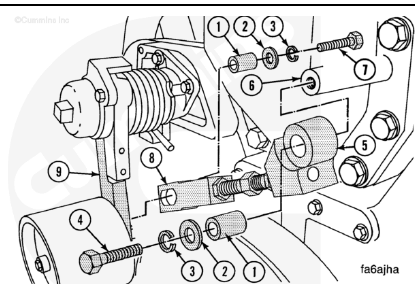
Install the lock washer and capscrew.

Torque Value: 45 n•m [35 ft-lb]

Slowly turn the wrench until the idler is against the belt.

Note : The fan hub pulley and the fan belt are shown removed for clarity.

When installing the solid control rod (turnbuckle) on older engines, the capscrews (4)(7) are 64 mm [2-1/2 in] in length. On the newer engines, the capscrew (4)(7) are 57 mm [2-1/4



in] in length. It is recommended that SAE Grade 8 capscrews that are 57 mm [2-1/4 in] be installed or the capscrews can break.

Install a spacer (1), a heavy flat washer (2), and lock washer (3).

Install a SAE Grade 8 capscrew (4), 57 mm [7/16-14 x 2-1/4 in] in the upper control rod end (5).

Hand tighten the capscrew.

Install the upper control rod end in the fan hub support (6).

Install a spacer (1), a lock washer (3), and heavy flat washer (2).

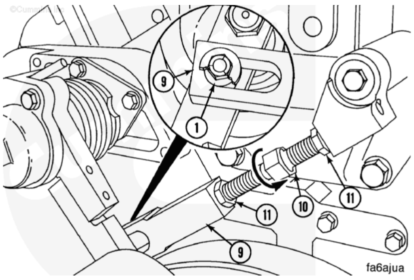
Install a SAE Grade 8 capscrew (7), 57 mm [7/16-14 x 2-1/4 in] in the lower control rod end (8).

Install the lower control rod end on the idler arm (9).

Tighten the capscrews (4)(7).

Torque Value: 90 n•m [65 ft-lb]

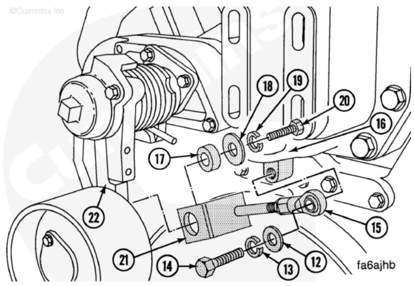
Note : The fan belt must be installed and under the tension of the fan idler arm spring to adjust the control rod. The fan belt and a portion of the flat washer are not shown for clarity.



Turn the adjusting screw (10) until the end of the slot on the lower control rod end (9) is touching the spacer (1). One of the nuts has left handed threads.

Hold the adjusting screw and tighten the two jam nuts (11).

To install the control rod with spring, install the flat washer (12), lock washer (13), and capscrews (14) in the upper end of the control rod (15).



Install the control rod in the fan support (16).

Tighten the capscrew.

Torque Value: 60 n•m [45 ft-lb]

Install the spacer bushing (17), flat washer (18), lock washer (19), and capscrew (20) in the lower end of the control rod (21).

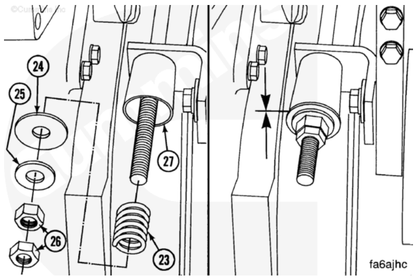
Install the lower end of the control rod on the fan idler are (22).

Tighten the capscrews.

Torque Value: 60 n•m [45 ft-lb]

⚠ CAUTION ⚠

Do not tighten the inner fan nut excessively. If the jam nut is to tight, the spring retainer will bend and the control rod will fail.



Install the following parts:

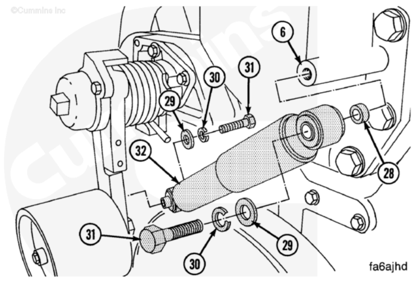
- Spring (23)
- Spring retainer washer (24)
- Flat washer (25)
- Jam nuts (quantity 2)(26).

Turn the inner jam nut until the spring retainer washer (24) touches the cylinder on the lower control rod end (27).

Hold the inner jam nut and tighten the outer jam nut.

⚠ CAUTION ⚠

The shock absorber must be installed with the larger outer tube of the shock absorber attached to the fan hub support. If the absorber is installed wrong, dirt can enter the tube and cause the part to fail.



Install the following parts to install the shock absorber:

- Spacer (28)
- Flat washer (29)
- Lock washer (30)
- Capscrew (31)
- Shock absorber (32).

Install the shock absorber (32) in the fan support (6).

Install the flat washer (29), lock washer (30), and capscrew (31) in the lower end of the shock absorber.

Install the shock absorber on the fan idler arm.

Tighten the two capscrews (31).

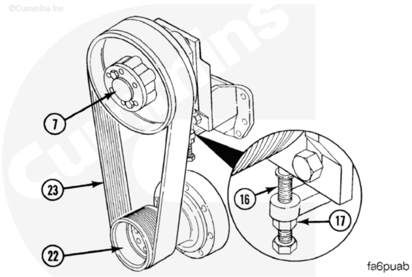
Torque Value: 60 n•m [45 ft-lb]

The following step is for installing the two pulley fan drive belt center 711 mm [28 in].

Install the poly vee 20 rib L-section fan belt (23) on the crankshaft pulley (22) and the fan hub pulley (7). Align the grooves in the belt on the ribs in the pulleys.

Make sure the heavy nut (17) is positioned to allow the adjusting capscrew (16) to turn freely.

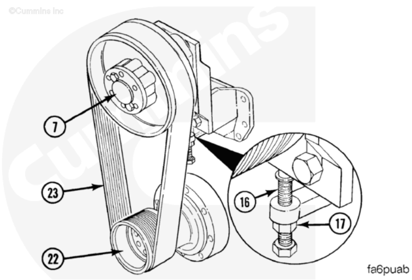
Turn the adjusting capscrew (16) **counterclockwise** to remove the slack from the belt.



The following step is for installing the two pulley fan drive belt centers 508mm [20 in], 559mm [22 in], and 610 mm [24 in].

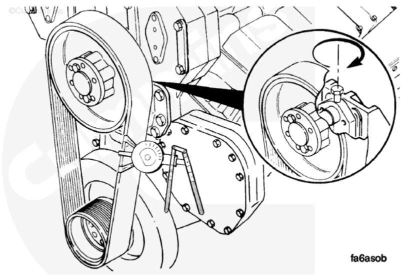
Install the poly vee 20 rib L-section fan belt (18) on the crankshaft pulley (17) and the fan hub pulley (7). Align the grooves in the belt on the ribs in the pulleys.

Turn the adjusting capscrew (10) **clockwise** to remove the slack from the belt.



Adjust

Only one method is acceptable for setting the two pulley fan drive belt tension. The recommended method is to use a belt tensioner gauge.

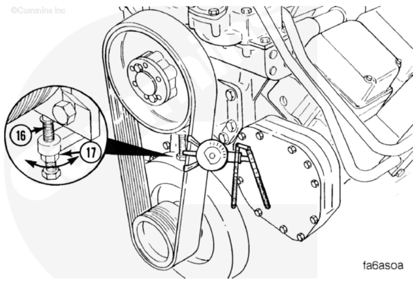


⚠ WARNING ⚠

The fan belt idler is under tension. Do not allow hands to get between the idler, fan, or the fan hub. Failure to do so can result in personal injury.

⚠ CAUTION ⚠

Incorrect belt tension can cause component failure.



Install the belt tension gauge, Part Number 3823772, or equivalent, on the belt in the middle between the two pulleys. Continue tightening the adjusting capscrew to a belt tension of

2668.9 to 2891.3 N [600 to 650 lbf] The belt tension will increase when the capscrews tighten the fan hub assembly to the fan support.

Tighten the capscrews.

Torque Value: 285 n•m [210 ft-lb]

Remove the tension gauge and position the gauge on the other side of the belt. Check to make sure the tension is correct 2891.3 to 3336.2 N [650 to 750 lbf]. If the belt tension is **not** correct, loosen the capscrews and adjust to the correct tension again.

Torque Value: 285 n•m [210 ft-lb]

Last Modified: 31-Jan-2006